

Practice Problems

Equations used:

Vibrational states in molecules:

$$E_N = (N + \frac{1}{2})\hbar\omega; \omega = \sqrt{\frac{k}{m}} = 2\pi f$$

Use harmonic mean for mass

$$m = \left(\sum \frac{1}{m_i} \right)^{-1} \left[= \frac{m_1 \cdot m_2}{m_1 + m_2} \right]$$

Rotational energy also exists, from Q.N. L

$$E_L = \frac{L(L+1)\hbar^2}{2mR_{eq}^2}$$

For dilute gas, go straight to Maxwell-Boltzmann.

$$K_{avg} = \frac{3}{2}kT$$

$$k = 1.38 \times 10^{-23} \text{ J/K} = 8.617 \times 10^{-5} \text{ eV/K}$$

$$N(E) = \frac{2N}{\sqrt{\pi}(kT)^{3/2}} E^{1/2} e^{E/kT}$$

Approximate tiny distribution integrals.

$$\int_{E_m}^{E_m + \Delta E} N(E) dE \approx N(E) * \Delta E$$

Problem 1

What atoms have the ground state configurations: (a) $1s^2 2s^2 2p^1$, (b) $1s^2 2s^2 2p^6 3s^1$, (c) $[Ne] 3s^2 3p^5$

Solution

- (a) Boron (B)
- (b) Sodium (Na)
- (c) Chlorine (Cl)

Problem 2

What is the ground state configuration of the ion K^+ ?

Solution

$$1s^2 2s^2 2p^6 3s^2 3p^6$$

Problem 3

What is the ground state configuration of the ion S^{-2} ?

Solution

$$1s^2 2s^2 2p^6 3s^2 3p^6$$

Problem 4

What is the electron configuration of the first excited state of He?

Solution

$$1s^1 2s^1$$

Problem 5

For the hydrogen atom in the 3d excited state, give the possible values of (ℓ, m_ℓ, s, m_s) for the atom.

Solution

$$\ell = 2; m_\ell \in \{-2, -1, 0, 1, 2\}; s = \frac{1}{2}; m_s \in \{\frac{1}{2}, -\frac{1}{2}\}$$

Problem 6

List the quantum numbers (n, ℓ, m_ℓ, m_s) for all the electrons in a ground state nitrogen atom.

Solution

$$\begin{array}{l|ll} 1s & (1, 0, 0, -\frac{1}{2}) & (1, 0, 0, \frac{1}{2}) \\ 2s & (2, 0, 0, -\frac{1}{2}) & (2, 0, 0, \frac{1}{2}) \\ 2p & (2, 1, -1, \frac{1}{2}) & (2, 1, 0, \frac{1}{2}) \quad (2, 1, 1, \frac{1}{2}) \end{array}$$

Problem 7

Infrared photons of wavelength 3440 nm are observed in the emission spectrum of molecular HCl. You suspect that this is from excitations of vibrational modes of the molecule as a harmonic oscillator. The mass of H is 1.008 u and the mass of Cl is 34.97 u where the atomic mass unit $u = 1.63 \times 10^{-27}$ kg. What is the ‘spring constant’ of this ionic bond?

Solution

Energy of a photon can be computed from wavelength, so let's do that.

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{3440 \text{ nm}} = 0.36 \text{ eV} \times \frac{1.602 \times 10^{-19} \text{ J}}{1 \text{ eV}} = 5.7672 \times 10^{-20} \text{ J}$$

This is the difference between two vibrational nodes' energies, or equivalent to $\hbar\omega$. Since $\omega = \sqrt{\frac{k}{m}}$, we can from here calculate the spring constant k . Make sure to use the harmonic mean for the mass.

$$\begin{aligned} E &= \hbar\omega = \hbar\sqrt{\frac{k}{m}} \\ m &= \frac{m_1 * m_2}{m_1 + m_2} = \frac{34.97 * 1.008}{34.97 + 1.008} = 0.9798 \\ k &= \frac{E^2 m}{\hbar^2} = \frac{(5.7672 \times 10^{-20} \text{ J})^2 (0.9798 \times 1.63 \times 10^{-27} \text{ kg})}{(1.054 \times 10^{-34} \text{ J} \cdot \text{s})^2} = \boxed{478 \text{ N/m}} \end{aligned}$$

Problem 8

Consider the diatomic molecule H_2 . You can model it as two point masses (each of mass 1.08 u where $1u = 931.5 \text{ MeV}/c^2$) with a separation of 0.074 nm. What is the wavelength of the photons emitted when this ‘quantum rotor’ transitions with the lowest possible energy change?

Solution

The minimum transition would come from transition between $L = 1$ to $L = 0$. We can calculate the energy transition here. Don’t forget that we use the harmonic mean for the mass.

$$\begin{aligned} E &= \frac{L(L+1)\hbar^2}{2mR_{eq}^2} = \frac{1(2)(hc)^2}{(2\pi)^2 2mR_{eq}^2 c^2} = \frac{2(1240 \text{ eV} \cdot \text{nm})^2}{(2\pi)^2 (931.5 \text{ MeV})(0.074 \text{ nm})^2} \\ &= \frac{2 * 1.5376 \text{ eV}}{4\pi^2 931.5 * 5.476 \times 10^{-3}} = \frac{3.0752 \text{ eV}}{64.1} = 0.048 \text{ eV} \\ \lambda &= \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{0.048 \text{ eV}} = \boxed{81.2 \mu\text{m}} \end{aligned}$$

Problem 9

Consider 1 mol of a dilute He gas at temperature 293 K. (a) What is the mean value of the energy, E_m , of an atom? (b) What fraction of the atoms has an energy in an interval of $0.01 * E_m$, centered at E_m ?

Solution (a)

Use the Maxwell-Boltzmann energy distribution and find the average kinetic energy.

$$K_{avg} = \frac{3}{2}kT = \frac{3}{2} * 8.617 \times 10^{-5} \text{ eV/K} * 293 \text{ K} = \boxed{37.87 \times 10^{-3} \text{ eV}}$$

Solution (b)

Recall the Maxwell-Boltzmann distribution.

$$N(E) = \frac{2N}{\sqrt{\pi}(kT)^{3/2}} E^{1/2} e^{E/kT}$$

Divide both sides by the number of total particles, which happens to already be in the equation.

$$\frac{N(E)}{N} = \frac{2}{\sqrt{\pi}(kT)^{3/2}} E^{1/2} e^{E/kT}$$

Multiply both by dE and take the integral over the span. Use the approximation.

$$\begin{aligned} \int_{0.995 E_m}^{1.005 E_m} \frac{N(E)}{N} dE &\approx \frac{2}{\sqrt{\pi}(kT)^{3/2}} E_m^{1/2} e^{E_m/kT} * \Delta E \\ &= \frac{2}{\sqrt{\pi}(25.25 \times 10^{-3})^{3/2}} (37.87 \times 10^{-3})^{3/2} e^{-\frac{37.87 \times 10^{-3}}{25.25 \times 10^{-3}}} * 0.01 \\ &= \frac{32.89 \times 10^{-6}}{7.11 \times 10^{-3}} = 0.0046 \end{aligned}$$

This means about $\boxed{0.46\%}$ of all atoms are in that range.

Problem 10

The speed of light is the fastest speed that any signal can travel. We say that if two events (A and B) in space time can be connected by a signal that travels at or less than the speed of light then these two events can be causally connected. That is, the earlier event can cause the later event and all possible different observer frames will agree on which event was the earlier event. (Effects can not occur before causes)

(a) Show that if A and B are connected by a light signal in a space-time frame where A happens first, then no matter what other frame observes the events, B always happens later than A.

But if two events can only be connected by a signal that travels faster than light, which is impossible, then the two events cannot be causally connected- and there can be an observer frame that observes the events to happen simultaneously, a different observer frame observes event A to happen first and event B later; and another observer frame to observe that the event B happens first.

(b) Pick two events A and B that would be connected by a signal that travels at twice the speed of light and pick A to have happened first and list them.

(c) What is the relative velocity of a frame where observers in that frame would see A and B happen at the same time?

(d) What are the relative velocities of frames that would see B happen first?

Solution (a)

If event A causes event B by a light signal, then A and B are colinear in a spacetime diagram along a line of slope 1. A would also take place before B.

$$t_B > t_A$$

Suppose for contradiction that in some frame of reference, event B takes place at the same time or before event A.

$$t'_B \leq t'_A$$

If we are in a relativistic frame of reference, we are traveling at a specific speed u . We can apply the Lorentz transformation for the time here.

$$\begin{aligned} t' &= \frac{t - (u/c^2)x}{\sqrt{1 - u^2/c^2}} \\ \frac{t_B - (u/c^2)x_B}{\sqrt{1 - u^2/c^2}} &\leq \frac{t_A - (u/c^2)x_A}{\sqrt{1 - u^2/c^2}} \\ t_B - (u/c^2)x_B &\leq t_A - (u/c^2)x_A \end{aligned}$$

Problem 11

Consider two protons headed towards each other with the same speeds. Each proton has mass $m_p = 938.8 \text{ eV}/c^2$. They collide and create a deuteron with mass $m_d = 1875.6 \text{ MeV}/c^2$ and a positively charged particle called the 'pion plus'. The pion particle symbol is π^+ and its mass is $m_\pi = 139.6 \text{ MeV}/c^2$. This interaction can be symbolized with $p^+ + p^+ \rightarrow \pi^+ + d^+$.

(a) What is the minimum kinetic energy that each proton must have in order for this to be able to happen?

(b) Say that the two protons head into the collision from opposite directions each with 100 MeV of kinetic energy. How much momentum will the pion have after it is created?

Chapter 8

Many Electron Atoms

Equations used:

$$Z_{eff} = q_{\text{nucleus}} - q_{\text{screening electrons}} \quad (8.1)$$

$$E_n = (-13.6 \text{ eV}) \frac{Z_{eff}^2}{n^2} \quad (8.2)$$

For transition energies:

$$\Delta E = 13.6 \text{ eV} (Z - 1)^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \quad (8.3)$$

For electrons in incomplete shells, exists angular (L) & magnetic (M_L) Q.N's:

$$|L_i| = \hbar \sqrt{l_i(l_i + 1)} \quad (8.4)$$

$$L_{max} = l_1 + l_2; L_{min} = |l_1 - l_2| \quad (8.5)$$

$$L \in [L_{min}, L_{max}] \cap \mathbb{N} \quad (8.6)$$

$$M_L = \sum m_{l,i}; M_L \in [-L, L] \cap \mathbb{N} \quad (8.7)$$

S can be computed same as M_L but with spin quantum number

Hund's rule: $L = M_{L,max}$, $S = M_{S,max}$;

maximize total spin

$\Delta L \in \{0, \pm 1\}; \Delta S = 0$

HeNe laser produces photons

$$\lambda = 632.8 \text{ nm}; E = 1.96 \text{ eV} \quad (8.8)$$

$$E_0 = \sqrt{\frac{2I}{c\epsilon_0}}; I = \frac{P}{A} \quad (8.9)$$

8.1 Problem 1

(a) List the six possible sets of quantum numbers (n, l, m_l, m_s) of a 2p electron. (b) Suppose we have an atom such as carbon, which has two 2p electrons. Ignoring the Pauli principle, how many different possible combinations of quantum numbers of the two electrons are there? (c) How many of the possible combinations of part (b) are eliminated by applying the Pauli principle? (d) Suppose carbon is in an excited state with configuration $2p^1 3p^1$. Does the Pauli principle restrict the choice of quantum numbers for the electrons? How many different sets of quantum numbers are possible for the two electrons?

8.1.1 Solution (a)

$$\begin{array}{lll} (2, 1, -1, -\frac{1}{2}) & (2, 1, 0, -\frac{1}{2}) & (2, 1, 1, -\frac{1}{2}) \\ (2, 1, -1, \frac{1}{2}) & (2, 1, 0, \frac{1}{2}) & (2, 1, 1, \frac{1}{2}) \end{array}$$

8.1.2 Solution (b)

Take the triangle number of six (for the six configurations).

$$T_6 = \frac{n(n+1)}{2} = \frac{6 * 7}{2} = \boxed{21} \quad (8.10)$$

8.1.3 Solution (c)

Only six, since there are six states and Pauli only excludes them when the two states are identical. This totals us at 15 states.

8.1.4 Solution (d)

The Pauli principle still applies, but it would not be important here. The two have different values for quantum number n . Furthermore, we would no longer have to account for duplicates, so we can just multiply the number of possible combinations in each shell. This gives us $6 * 6 = \boxed{36}$ possible combinations.

8.2 Problem 3

- (a) How many different sets of quantum numbers (n, l, m_l, m_s) are possible for an electron in the 4f level?
- (b) Suppose a certain atom has three electrons in the 4f level. What is the maximum possible value of the total m_s of the three electrons?
- (c) What is the maximum possible total m_l of three 4f electrons?
- (d) Suppose an atom has ten electrons in the 4f level. What is the maximum possible value of the total m_s of the ten 4f electrons?
- (e) What is the maximum possible total m_l of ten 4f electrons?

8.2.1 Solution (a)

$$2(2\ell + 1) = 2(2 * 3 + 1) = 2 * 7 = \boxed{14} \quad (8.11)$$

8.2.2 Solution (b)

Each of them could have a value of $m_s = \frac{1}{2}$. Multiplying that by three, we get $\underline{\frac{3}{2} = 1.5}$

8.2.3 Solution (c)

They would have m_l values of 3, 3, and 2 first. The total would be $\boxed{8}$.

8.2.4 Solution (d)

Start by filling it with seven electrons where $m_s = \frac{1}{2}$. This leaves us with a total m_s of $\frac{7}{2}$ and three electrons remaining. All three of these remaining electrons would have to have $m_s = -\frac{1}{2}$. Adding the two together, the total value would be $\boxed{2}$.

8.2.5 Solution (e)

Two electrons would have values of 3, two of 2, two of 1, etc. all the way down to and including -1.

$$2(3 + 2 + 1 + 0 - 1) = 2 * 5 = \boxed{10} \quad (8.12)$$

8.3 Problem 5

- (a) List all elements with a p^3 configuration.
- (b) List all elements with a d^7 configuration.

8.3.1 Solution (a)

Nitrogen, Phosphorus, Arsenic, Antimony, and Bismuth, plus whatever has atomic number 115.

8.3.2 Solution (b)

Cobalt, Rhodium, Iridium, and Meitnerium.

8.4 Problem 7

- (a) What is the electronic configuration of Fe?
- (b) In its ground state, what is the maximum possible total m_s of its electrons?
- (c) When the electrons have their maximum possible total m_s , what is the maximum total m_l ?
- (d) Suppose one of the d electrons is excited to the next highest level. What is the maximum possible total m_s , and when m_s has its maximum total what is the maximum total m_l ?

8.4.1 Solution (a)

$$\boxed{1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6}$$

8.4.2 Solution (b)

Anything but the outermost shell ($3d$) would be empty and cancel out. At 6 electrons, we would have at most five electrons with $m_s = \frac{1}{2}$ and at least one electron with $m_s = -\frac{1}{2}$. This has a total m_s of $\boxed{2}$.

8.4.3 Solution (c)

Subshells with $m_l \in \{+2\}$ would be full (two electrons). Subshells with $m_l \in \{+1, 0, -1, -2\}$ would have only one electron.

$$2 * 2 + (1 + 0 - 1 - 2) = 4 - 2 = \boxed{2} \quad (8.13)$$

8.4.4 Solution (d)

To achieve the maximum value of m_s , we would have to take our answer from (b) move an electron from $m_s = -\frac{1}{2}$ to $m_s = \frac{1}{2}$. This would have the total value of $m_s = \boxed{3}$. The next energy level that would be accessible by an electron would be $4p$, so the maximum possible value of m_l would be 1. Add everything together and make the replacement.

$$2 + 1 + 1 + 0 - 1 - 2 = \boxed{1} \quad (8.14)$$

8.5 Problem 9

The ground state of neutral beryllium ($Z = 4$) is $1s^2 2s^2$. Use the electron screening model to predict the energies of the following excited states: (a) $1s^2 2s^1 3p^1$ (measured -2.02 eV) and (b) $1s^2 2s^1 4d^1$ (-0.90 eV).

8.5.1 Solution (a)

Here, we use equation 8.1.

$$E_n = (-13.6 \text{ eV}) \frac{Z_{eff}^2}{n^2} = (-13.6 \text{ eV}) \frac{(4-3)^2}{3^2} = (-13.6 \text{ eV}) \frac{1}{9} = \boxed{-1.5 \text{ eV}} \quad (8.15)$$

8.5.2 Solution (b)

$$E_n = (-13.6 \text{ eV}) \frac{Z_{eff}^2}{n^2} = (-13.6 \text{ eV}) \frac{(4-3)^2}{4^2} = (-13.6 \text{ eV}) \frac{1}{16} = \boxed{-0.85 \text{ eV}} \quad (8.16)$$

8.6 Problem 11

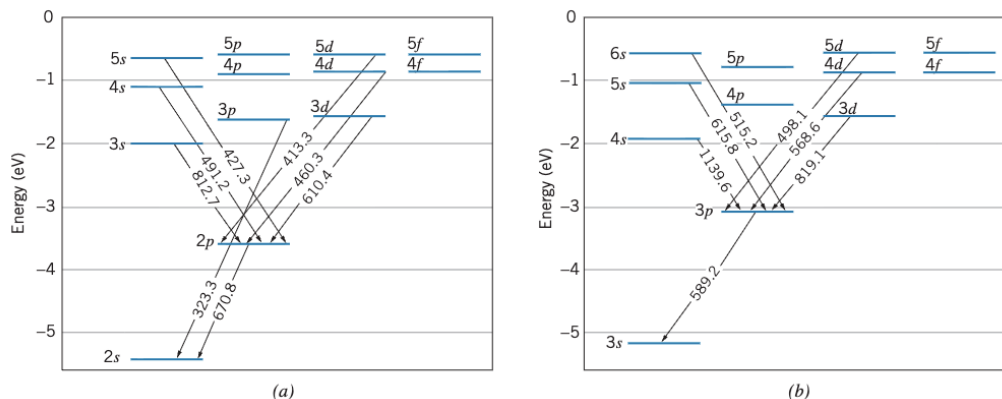


Figure 8.4

- Using the information for lithium given in Figure 8.4, compute the energy difference of the 3p and 3d states.
- Compute the energy of the 3s, 4s, and 5s states above the ground state.
- The ionization energy of lithium in its ground state is 5.39 eV. What is the ionization energy of the 2p state? Of the 3s state?

8.6.1 Solution (a)

3p to 2s emits a photon of wavelength $\lambda = 323.3 \text{ nm}$. 3d to 2p emits a photon of wavelength $\lambda = 610.4 \text{ nm}$. 2p to 2s emits a photon of wavelength $\lambda = 670.8 \text{ nm}$. From here, make a pathway from 3d to 3p, ensuring to make everything in terms of energy.

$$E_{3d \rightarrow 3p} = E_{3d \rightarrow 2p} + E_{2p \rightarrow 2s} - E_{3p \rightarrow 2s} = hc \left(\frac{1}{\lambda_{3d \rightarrow 2p}} + \frac{1}{\lambda_{2p \rightarrow 2s}} - \frac{1}{\lambda_{3p \rightarrow 2s}} \right) \quad (8.17)$$

$$= (1240 \text{ eV} \cdot \text{nm}) \left(\frac{1}{610.4 \text{ nm}} + \frac{1}{670.8 \text{ nm}} - \frac{1}{323.3 \text{ nm}} \right) \quad (8.18)$$

$$= (1240 \text{ eV}) (1.64 \times 10^{-3} + 1.49 \times 10^{-3} - 3.09 \times 10^{-3}) \quad (8.19)$$

$$= 0.04 * 1240 \text{ eV} = \boxed{49.6 \text{ eV}} \quad (8.20)$$

8.6.2 Solution (b)

Assume

8.7 Problem 13

Compute the K_α X ray energies of calcium ($Z = 20$), zirconium ($Z = 40$), and mercury ($Z = 80$). Compare with the measured values of 3.69 keV, 15.8 keV, and 70.8 keV. (See Question 16).

8.7.1 Solution

Calcium: K_α is the transition between $n = 2$ and $n = 1$.

$$\Delta E = (13.6 \text{ eV})(Z - 1)^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) = (13.6 \text{ eV})19^2 \left(\frac{1}{2^2} - \frac{1}{1^2} \right) = \boxed{3.6822 \text{ keV}} \quad (8.21)$$

This is not far off from the measured value.

Zirconium:

$$\Delta E = (13.6 \text{ eV})(Z - 1)^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) = (13.6 \text{ eV})39^2 \left(\frac{1}{2^2} - \frac{1}{1^2} \right) = \boxed{15.5142 \text{ keV}} \quad (8.22)$$

This is not far off from the measured value.

Mercury:

$$\Delta E = (13.6 \text{ eV})(Z - 1)^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) = (13.6 \text{ eV})79^2 \left(\frac{1}{2^2} - \frac{1}{1^2} \right) = \boxed{63.6582 \text{ keV}} \quad (8.23)$$

This is much farther off from the measured value.

8.8 Problem 15

Use Hund's rules to find the ground-state L and S of (a) Ce, configuration $[\text{Xe}]6s^24f^15d^1$; (b) Gd, configuration $[\text{Xe}]6s^24f^75d^1$; (c) Pt, configuration $[\text{Xe}]6s^14f^{14}5d^9$.

8.8.1 Solution (a)

Hund's rule says that $L = M_{L,max}$ and $S = M_{S,max}$.

The electrons in the outermost shells are $4f^15d^1$. The maximum possible values for m_l are 3 for the f -orbital and 2 for the d -orbital. Together, they total $L = M_{L,max} = 3 + 2 = \boxed{5}$.

The electrons in the outermost shells are $4f^15d^1$. The maximum possible values for m_s are $\frac{1}{2}$ both. Together, they total $S = M_{S,max} = \frac{1}{2} + \frac{1}{2} = \boxed{1}$.

8.8.2 Solution (b)

The electrons in the outermost shells are $4f^75d^1$. The maximum possible values for m_l are two copies of 3, 2, and 1 and one copy of 0 from the $4f$ orbital, and one copy of 2 for the d -orbital. Together, they total $L = M_{L,max} = 2(3 + 2 + 1) + 2 + 0 = \boxed{14}$.

The electrons in the outermost shells are $4f^75d^1$. At most, all 7 electrons in the $4f$ orbital would have value $\frac{1}{2}$, as with the one electron in the $5d$ orbital. Together, they total $S = M_{S,max} = 8 * \frac{1}{2} = \boxed{4}$.

8.8.3 Solution (c)

Here, the $4f$ -orbital is full, so we can disregard that. In the $6s$ orbital, the maximum possible value of m_l is just 0. In the $5d$ orbital, the maximum possible value of M_L would come when only one value of m_l is not taken up, being -2 , which would make the total sum $L = \boxed{2}$.

For a similar reason, the maximum possible value of M_S is $S = \boxed{1}$.

8.9 Problem 17

A certain excited state of an atom has the configuration $4d^1 5d^1$. What are the possible L and S values?

8.10 Problem 19

A small helium-neon laser produces a light beam with an average power of 3.5 mW and a diameter of 2.4 mm. (a) How many photons per second are emitted by the laser? (b) What is the amplitude of the electric field of the light wave? Compare this result with the electric field at a distance of 1 m from an incandescent light bulb that emits 100 W of visible light.

8.10.1 Solution (a)

A watt is equivalent to a joule per second. Since this is one second only, we can just multiply the power by one second to get the energy from one second to be $E = 3.5 \text{ mJ}$. Each HeNe laser photon has an energy of 1.96 eV, or $3.13 \times 10^{-19} \text{ J}$. Dividing the energy delivered by the energy of each photon, we get $\boxed{1.11 \times 10^{16}}$ photons.

8.10.2 Solution (b)

First find the intensity.

$$I = \frac{P}{A} = \frac{3.5 \text{ mW}}{\pi(1.2 \text{ mm})^2} = 773.6 \text{ W/m}^2 \quad (8.24)$$

$$E_0 = \sqrt{\frac{2I}{c\epsilon_0}} = \sqrt{\frac{2 * 773.6 \text{ W/m}^2}{2.998 \times 10^8 \text{ m/s} * 8.85 \times 10^{-12} \text{ C}^2/(\text{Nm}^2)}} \quad (8.25)$$

$$= \sqrt{\frac{1547.2}{2.65 \times 10^{-3}}} \text{ N/C} = \sqrt{583900} \text{ N/C} = \boxed{764 \text{ N/C}} \quad (8.26)$$